WEEK 1 · DAY 3 · RULE BOOK

Calm Down the Noise: Rolling Windows

Follow it step by step. Every common error and its fix is here.

Your goal today: smooth the daily NOC ticket series with a rolling window, find out what smoothing hides, and earn **14 green ✅ ticks**. Time: about 60 minutes. Keep this page open next to Colab.

Before you start (2 minutes)

- ☐ Open the **W1D3-exercise** notebook with the **Open in Colab** button.
- ☐ **File → Save a copy in Drive.** Work only in the new tab.
- ☐ Run **STEP 0** (Shift + Enter). Wait for ✅ `Ready!`

The 7 golden rules for today

#	Rule	Why
1	**Run cells top to bottom**	Every cell uses what the cells above made (`tk`, `roll7`, `margin`, …)
2	**Roll one column, not the whole table**: `tk["tickets_raised"].rolling(7)`	The column is the series. The table has other columns too
3	**A window is a number of rows**: `rolling(7)` on daily data = 7 days	On hourly data, `rolling(24)` = one day
4	**The first N−1 values are `NaN`**	A 7-day average doesn't exist until day 7. It's not an error
5	**Trailing (the default) for anything live**	`center=True` uses future days: reports only
6	**Never report an SLA from a smoothed line**	Short outages disappear inside the average
7	**`True` / `False` have no quotes; text does**	`CENTER = True`, but `"half"`, `"no"`, `"high"`

The exercise, step by step

Task	What you do	The tool you need
A1	Name the time column	`"date"`

Task	What you do	The tool you need
A2	Weekend vs weekday volume	read the two averages: `"a quarter"`, `"half"` or `"the same"`
B1	Window size for one week	`7`
B2	Count the empty rows at the start	`tk["roll7"].isna().sum()`
B3	Which window still zig-zags?	read the chart: `3`, `7` or `30`
B4	Prove 7 cancels the weekend	`roll7_by_day.max() - roll7_by_day.min()`
B5	Make a centred window	`CENTER = True`
B6	Is the centred line usable live?	`"yes"` or `"no"`
B7	Count weekly buckets	`len(weekly)`
B8	Which alpha follows the data closely?	`"high"` or `"low"`
B9	Share of days with zero P1 tickets	`(tk["priority_1"] == 0).mean()`
B10	Outage hours: raw vs smoothed	`(margin < 0).sum()`, `(smooth < 0).sum()`
B11	Weekly mean + "investigate 38 → 41?"	`wk["tickets_raised"].mean()`, `"yes"` / `"no"`
🌶️ **Bonus**	Count the odd weeks	`((q - q.mean()).abs() > 2 * q.std()).sum()`

The smoothing toolbox

```
s = tk["tickets_raised"]

s.rolling(7).mean()                # trailing 7-day average (safe for live use)
s.rolling(7).std()                 # how wobbly the last 7 days were
s.rolling(30, center=True).mean()  # centred: uses 15 future days (reports only!)
s.ewm(alpha=0.3).mean()            # exponential smoothing: high alpha = fast, low =
smooth
s.resample("W").sum()              # weekly buckets: one value per week (not a rolling
window)
```

Pick a window: daily tickets → start with **7**. Rare counts (P1) → **30**. Hourly data → **24** for a day.

When things go wrong: error → fix

You see	It means	Fix
`AssertionError: ✍️ Do A1 first…`	A blank above is still empty	Fill it in, run that cell, then run this one again
`NameError: name 'tk' is not defined`	Colab forgot, or you skipped a cell	Run STEP 0 and the cells above, top to bottom
`ValueError: window must be an integer`	`rolling()` got text or nothing	`WINDOW = 7`, no quotes
`KeyError: 'roll7'`	You skipped the cell that makes `roll7`	Run the B1 cell after filling `WINDOW`
`TypeError: unsupported operand…`	You did maths with text (e.g. `"7"`)	Numbers have no quotes
A ✅ check says ❌ for `CENTER`	You typed `"True"` with quotes	`CENTER = True`
Chart is flat or empty	The cell above it didn't run	Re-run the cells in order
Everything is weird	–	**Runtime → Restart session and run all**

Stuck? Climb the ladder

1. **Read the ❌ hint**: it usually says exactly what's wrong.
2. **Check this rule book's error table.**
3. **Ask your neighbour.**
4. **Ask the AI button** to *explain the error*. Paste your code and the error, never real data.
5. **Raise your hand.**

🏁 **Finished?** Run `scoreboard()`. 14 of 14 = 🏆. Then help someone near you.